Plant Soil (2013) 366:433–451
DOI 10.1007/s11104-012-1426-x

REGULAR ARTICLE

Modelling of Cd, Cu, Pb and Zn transport in metal contaminated soil and their uptake by willow (*Salix* × *smithiana*) using HYDRUS-2D program

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Received: 20 April 2012 / Accepted: 14 August 2012 / Published online: 9 September 2012

Abstract
Aims The main aim of this study was to validate the HYDRUS-2D model for the simulation of Cd, Cu, Pb and Zn transport within a soil column and their accumulation by willows.
Methods A simulation of metal transport and uptake by willow was implemented using the HYDRUS-2D code. Two scenarios of the column experiment were compared: soil (C) and soil with planted willows (V). Seeping water, soil water content and actual transpiration were measured. Metal contents in soil water and willows were analysed.
Results The single-porosity model (applied for isotropic soil media) was sufficient for scenario C. The single-porosity (applied for anisotropic soil media) and dual-porosity models (characterizing non-equilibrium water flow) were explored in scenario V. Measured cumulative Cd and Zn uptake showed a 20- and 10-times higher accumulation, respectively, in comparison with the modelled ones. On the other hand, modelled cumulative Pb uptake was reduced by reducing the maximum value of the root uptake concentration c_{Root}.
Conclusions The HYDRUS-2D program was usable for modelling of Cd, Cu, Pb and Zn transport and their willow uptake. Additionally, consideration of dual-porosity soil media as well as anisotropy was suitable for the experiment with roots presence in the soil.

Keywords Soil column · HYDRUS-2D · Numerical optimization · Anisotropy and dual-porosity soil media · Metal transport · Root metal uptake · *Salix* × *smithiana*

Responsible Editor: Fangjie Zhao.

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Introduction

Phytoextraction is currently an extensively studied method for extracting metals (such as Cd, Cu, Pb and Zn) from polluted sites using different plant species. In metal contaminated soil, phytoextraction technologies are based preferably on the use of natural fast growing plant species such as willows (*Salix spp.*; Baum et al. 2006; Ruttens et al. 2011). In general, willows are suitable for phytoremediation of polluted sites mainly due to their high biomass production (Greger and Landberg 1999) and the possibility of frequent harvesting of aboveground parts (Riddell-Black 1993; Pulford and Watson 2003). Accumulation of metals into the willow tissues, which is necessary for successful phytoextraction, is individual for each specific metal. Willows are especially effective in the extraction of Cd and Zn from soil, and their subsequent transport to the aboveground biomass (Landberg and Greger 1996; Rosselli et al. 2003). There are many studies such as Dos Santos Utmazian and Wenzel (2007); Zimmer et al. (2009); Jensen et al. (2009), which confirm the suitability of different willow clones for Cd and Zn phytoextraction into the tissues. On the other hand, Cu and Pb are metals, which are strongly fixed in the soil mainly due to intensive sorption into organic matter (Zhou and Wong 2001; Komárek et al. 2009) and Al, Fe, Mn (hydr)oxides (Adriano 2001; Dijkstra et al. 2004). Hence, Cu and Pb uptake into the experimental willows as well as into the other plant species is reduced in comparison with Cd a Zn (Labrecque et al. 1995; Nissen and Lepp 1997). Specifically, the willow potential for Cd, Cu, Pb and Zn phytoextraction was evaluated by Jensen et al. (2009). This approach was also used by Trakal et al. (2011), where the phytoextraction potential was compared using a column experiment.

On the other hand, another modern approach to evaluation of the phytoextraction potential for selected metals is mathematical modelling. A large number of mechanistic models of varying degree, complexity and dimensionality have been developed during recent years to describe (quantify) water flow (de Rooij 2000; Köhne et al. 2009), metal transport in the unsaturated zone (Sayyad et al. 2010; Fang et al. 2011) and also the metal phytoextraction potential represented by root metal uptake (Puschenreiter et al. 2005; Javaux et al. 2008; Ebbs et al. 2009). Among others, the HYDRUS program was chosen for the study of water flow (Šimůnek et al. 2008), metal transport through the soil (Jacques et al. 2008; Nikodem et al. 2010) and mainly of the root metal uptake (Šimůnek and Hopmans 2009). This model was recently used to simulate the transport of Cd (Moradi et al. 2005), Cu (Bahaminyakamwe et al. 2006), and Zn, Cu and Pb (Ngoc et al. 2009) in the soil. Despite the several applications of the HYDRUS software packages for simulating metal transport through the soils, limited information exists in the literature about their utilization for simulating Cd, Cu, Pb and Zn transport in soils coupled with their root uptake. Moreover, there is no evidence about the Cd, Cu, Pb and Zn root uptake into experimental willows, which are suitable for effective phytoextraction represented by the root metal uptake.

Our main objective was therefore to evaluate the ability of the HYDRUS program to simulate root Cd, Cu, Pb and Zn uptake into the experimental willows (*Salix × smithiana*), which was experimentally studied in the laboratory soil columns by Trakal et al. (2011). The main aims of this study were: (i) to estimate hydraulic properties of soil with and without the willows using the numerical inversion; and (ii) to validate the model (calibrated for water flow) for the simulation of Cd, Cu, Pb and Zn transport within the soil column and their root uptake into the willows.

Methods and materials

Soil preparation and analyses

The experimental soil originated from a site which is one of the most polluted in Central Europe due to the atmospheric deposition of metals from a Pb smelter (Příbram, CZE, GPS: 49°42′N, 13°58′E). Soil samples were taken from the surface layer (0–25 cm) of a Gleyic Cambisol (planted with fast growing plants—poplars and willows) situated approximately 1 km NE from the smelter. More detailed information about this smelter and metals emissions is listed in Ettler et al. (2001, 2004). Soil samples were air-dried; sieved through a 2 mm sieve and homogenized to determine selected initial soil characteristics (Table 1).

Soil pH was measured using an inoLab Level 1 pH-meter in suspension using a 1:1.25 (w/v) ratio of soil, deionized water, and 0.01 M $CaCl_2$. Samples for soil CEC were prepared in a suspension of 1:50 (w/v) ratio of soil and 0.1 M $BaCl_2$ and consequently the CEC determination was carried out using ICP-OES (Varian, VistaPro, Australia). Dissolved organic carbon (DOC) was determined in extractions of 0.01 M $CaCl_2$, using a colorimetric method on (SKALAR plus SYSTEM apparatus, Nederland). The total organic carbon (TOC) was determined by wet oxidation with $K_2Cr_2O_7$ as well as by the measurement of the absorbance at 590 nm. Available nutrients (Ca^{2+}, K^+ and Mg^{2+}) were attained with the Mehlich III soil extraction procedure and measured using F-AAS (Varian 280FS, Australia). Major inorganic anions (F^-, Cl^-, PO_4^{3-}, $SO_4^{2-}$and NO_3^-) were determined by ICS 90 (Dionex, USA). Total metal (Cd, Cu, Pb and Zn) content in the soil was determined by using a mixture of concentrated HNO_3, HCl and HF

Table 1 Selected initial soil basic characteristics. More detailed soil characteristics are shown in Trakal et al. (2011, 2012)

Texture [$g kg^{-1}$]			pH [–]		CEC[a] [$mmol kg^{-1}$]		Total metal content [$mg kg^{-1}$]			
Clay	Silt	Sand	H_2O	$CaCl_2$			Cd	Cu	Pb	Zn
62.0	388	550	6.75	6.38	57.4	8.82	29.3	1724	231	
DOC[c] [$g kg^{-1}$]	TOC[b] [$g kg^{-1}$]	N[d] [$mg kg^{-1}$]	Available nutrients [$mg kg^{-1}$]							
			Ca^{2+}	K^+	Mg^{2+}	F^-	Cl^-	NO_3^-	PO_4^{3-}	SO_4^{2-}
1.05	27.5	206	2718	118	171	8.23	29.8	299	14.5	107

[a] Cation Exchange capacity

[b] Total organic carbon

[c] Dissolved organic carbon

[d] Total nitrogen content in the soil

acids; the extraction solution was then analysed by ICP-OES (Varian, VistaPro, Australia).

Column experiment

Experimental columns were situated under controlled conditions in a greenhouse. Ten cylindrical polyethylene containers (30 cm high and 20 cm diameter; Fig. 1) were prepared. The base of each container was sealed, except for a central drainage hole used for collecting seeping water into polyethylene bottles. A one-centimetre layer of gravel was placed at the bottom of each column to ensure even (unrestricted) gravitational drainage of water from the bottoms of the tested soil columns. Experimental soil was packed into the columns (depth of 25 cm) and covered by gravel (1 cm) to reduce evaporation. Five of the columns were also planted with experimental willows using a 20 cm long cutting for each column (i.e. two variants were investigated for the two model scenarios: Control–C and Vegetation–V). The willow clone (*Salix × smithiana*, Willd.; Vysloužilová et al. 2003) obtained from Silva Tarouca Research Centre for Landscape and Decorative Horticulture in Průhonice, was used for the column experiment as a species producing high biomass yields. Irrigation (by deionized water) was applied gravitationally, using drip emitters connected to individual Mariotte style reservoirs for each column. The experiment was performed over two vegetation periods.

During the first vegetation period, the metal (Cd, Cu, Pb and Zn) concentration in seeping water was continuously measured (using ICP-OES). At the end of the vegetation period, willows were harvested to analyse biomass production and metal contents in the aboveground biomass (leaves and stems). The aboveground biomass samples were decomposed using the dry ashing procedure in a mixture of oxidizing gases (O_2+O_3+NO_x) in an Apion Dry Mode Mineralizer (Tessek, CZ). Metal content in the plant digest was determined by ICP-OES. Detailed descriptions, as well as the results, are published elsewhere (Trakal et al. 2011).

Apart from measuring the metal concentration in seeping water during the second vegetation period other measurements had also taken place. Cumulative outflow from each container over a period of time (after each irrigation) and the soil water content were also measured (at the end of each day) using a 25 cm long FDR (frequency domain reflectometry) probe CS616 (Campbell Sci., UK) vertically situated in the area closed the middle of column (three of five columns for each scenario were monitored). Furthermore, the metal concentration in the soil water, which was extracted from the soil using polyethylene soil water suction cups (installed to a depth of 15 cm) at the beginning of the second vegetation period, was also measured (using ICP-OES; Table 2). At the end of the second period the experimental columns were disintegrated. Before, the 100-cm^3 soil samples were taken from two columns of each scenario (from the depths of 5, 10 and 20 cm) to determine the bulk density and saturated soil water content. The willow plants were harvested and metal contents in aboveground biomass (leaves, stems) and roots were measured. Separated plant parts were then analysed for total metal content in tissues (aboveground biomass and roots) as well as for aboveground biomass and roots production. For detailed information and results see the paper Trakal et al.

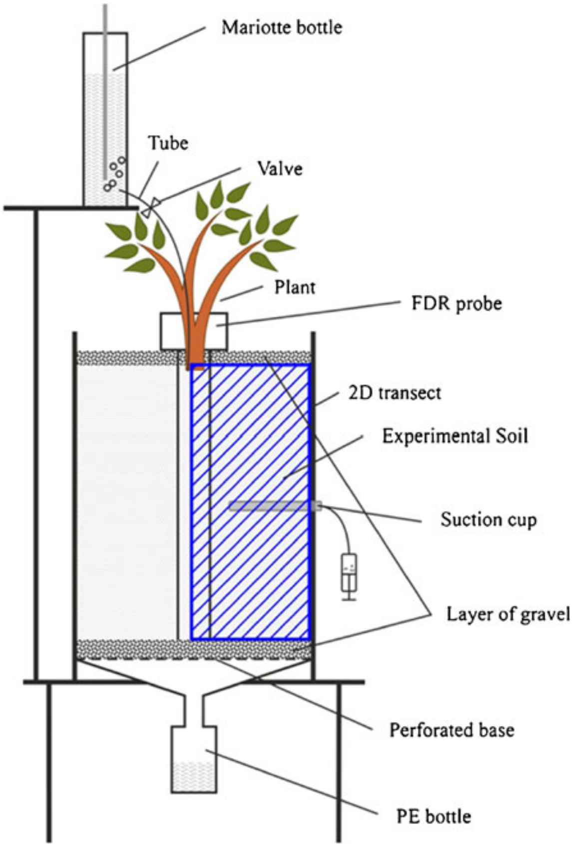

Fig. 1 Scheme of the soil column

(2011). During the final root separations from the soil columns, the homogenous root distributions in the soil columns were observed, which was assumed when calculating the total metal amount extracted from the soil where the undisturbed soil samples were taken. Roots were separated from the soil using: (i) careful soil disposal by different kinds of brushes; and (ii) consequent thorough washing by demineralized water. The root distribution within the column was assessed visually. The volume of roots was measured using the calibrated cylinder with water. Finally, Cd, Cu, Pb and Zn sorption behaviour in the experimental soil (which was at the end of the experiment either taken from the columns of C scenario or separated from roots in the column of V scenario) was also determined using a batch experiment (Poulsen and Hansen 2000; Trakal et al. 2012).

Mathematical modeling

Only the second vegetation period was used for the following numerical simulation using HYDRUS-2D. It was assumed that during the first vegetation period a homogenous root distribution in the soil column was established and the soil was compacted and structured (e.g., no changes in the root density distribution and soil hydraulic properties variability in time were expected).

Numerical model description

Water flow and metal transport in one column for each scenario was simulated using the single-porosity model implemented in HYDRUS-2D (Šimůnek et al. 2008). The Richards' Eq. 1, which describes isothermal water flow in the variably saturated anisotropic two-dimensional radially symmetric single-porosity system porous medium, was used:

$$\frac{\partial \theta}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rK(h)\left[K_{rj}^{A}\frac{\partial h}{\partial x_j} + K_{rz}^{A}\right]\right) + \frac{\partial}{\partial z}\left(K(h)\left[K_{zj}^{A}\frac{\partial h}{\partial x_j} + K_{zz}^{A}\right]\right) - S \quad (1)$$

Table 2 Metal concentration in soil water extracted from the soil column using the soil water suction cups at the beginning of the second vegetation period. Data shown are means±SD (n=5)

Metal [$nmol\,cm^{-3}$]	Control	Vegetation
Cd	0.09±0.05	0.05±0.02
Cu	0.37±0.19	0.29±0.14
Pb	0.62±0.20	0.61±0.23
Zn	0.71±0.35	0.67±0.27

where θ is the volumetric soil water content [L^3L^{-3}], which is a function of h, h is the pressure head [L], t is time [T], r is the radial coordinate [L], z is the vertical axis [L], and S is the sink term for root uptake [T^{-1}], $K(h)$ is the hydraulic conductivity function [LT^{-1}] and K_{ij}^{A} are components of a dimensionless anisotropy tensor K^A.

If an isotropic medium is assumed the diagonal components of the anisotropy tensor K^A are equal to 1 and off-diagonal components are equal to zero. If the directions of the anisotropy tensor K^A coincide with the global coordinate system (r, z), the off-diagonal entries are equal to 0 and the anisotropy tensor K^A can be written as follows:

$$K^A = \begin{vmatrix} K_{rr} & 0 \\ 0 & K_{zz} \end{vmatrix} \tag{2}$$

where $K_{rr}{}^A$, $K_{zz}{}^A$ are component associated with r and z coordinate direction, respectively.

The root water uptake in (1) was calculated assuming the Eq. 3 proposed by Feddes et al. (1978).

$$S = \alpha(h)b(z,r)T_pA \tag{3}$$

This equation calculates root water uptake from the potential plant transpiration multiplied by the alpha-function characterizing the plants ability to extract water from soil, which depends on the pressure head $\alpha(h)$ [−]. In addition, root water uptake depends on the root distribution within the soil $b(z,r)$ [L^{-3}], potential evapotranspiration T_p [LT^{-1}] and A [L^2] is surface area.

The soil hydraulic properties, soil water retention curve θ(h), and unsaturated hydraulic conductivity K (h), were described using the van Genuchten Eqs. 4, 5 (van Genuchten 1980):

$$\begin{aligned} &\theta_e = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r} = \frac{1}{(1 + |\alpha h|^n)^m} \ , h < 0 \\ &\theta_e = 1 \ , h \geq 0 \end{aligned} \tag{4}$$

$$\begin{aligned} &K(h) = K_s\theta_e^l\left[1 - \left(1 - \theta_e^{2/m}\right)^m\right]^2, \quad h < 0 \\ &K(h) = K_s, \quad h \geq 0 \end{aligned} \tag{5}$$

where θ_e is the effective soil water content [−], K_s is the saturated hydraulic conductivity [LT^{-1}], θ_r and θ_s are the residual and saturated soil water contents [L^3L^{-3}], respectively, l is the pore-connectivity parameter [−] (usually l=0.5), α is reciprocal of the air entry pressure [L^{-1}], n is related to the slope of the retention curve at the inflection point [−] and $m = 1 - 1/n$, $(n > 1)$.

The advection-dispersion Eq. 6 for metal transport in the soil was used in this model. The following simplified version describes metal transport, assuming only metal sorption on soil particles:

$$\frac{\partial \theta c}{\partial t} + \frac{\partial \rho_d s}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\theta D_r\frac{\partial c}{\partial r}\right) + \frac{\partial}{\partial z}\left(\theta D_z\frac{\partial c}{\partial z}\right) - \left(\frac{\partial qc}{\partial r} + \frac{\partial qc}{\partial z}\right) - S_c \tag{6}$$

where c [ML^{-3}] and s [MM^{-1}] are solute concentrations in the liquid and solid phases respectively, q is the volumetric water flux density [LT^{-1}], ρ_d is the soil bulk density [ML^{-3}], D_r and D_z are the dispersion coefficients [L^2T^{-1}]. The term S_c (= $S\,c$) describes the solute uptake by plants. The applied version of HYDRUS-2D allows simulating unreduced (i.e. all dissolved solute) solute uptake up to a set limited value c_{Root}. Only limited concentration is extracted by plant above this value in the simulation.

Assuming the equilibrium metal adsorption (Cd, Cu, Pb and Zn), the Freundlich Eq. 7 for describing adsorption isotherm was used (chosen for its higher efficiency compared with the Langmuir equation; Trakal et al. 2012), which describes the relationship between adsorbed concentration s and the liquid concentration c.

$$s = K_Fc^\beta \tag{7}$$

where K_F [$L^{3\beta}$ $M^{1-\beta}$ M^{-1}] and β [−] are empirical coefficients of the Freundlich equation.

The dual-porosity model in HYDRUS-2D was also used to describe possible non-equilibrium water flow and metal transport. In this case, the flow domain was divided into two domains: mobile domain (zones of flowing water) and immobile domain (zones of stagnant water). The Richards equation was applied to describe water flow in mobile zones and a mass balance equation was used to describe soil water content dynamic in immobile zones:

$$\begin{aligned} &\frac{\partial \theta_{mo}}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rK_{mo}(h_{mo})\frac{\partial h_{mo}}{\partial r}\right) + \frac{\partial}{\partial z}\left(K_{mo}(h_{mo})\frac{\partial h_{mo}}{\partial z} + K_{mo}(h_{mo})\right) - S_{mo} - \Gamma_w \\ &\frac{\partial \theta_{im}}{\partial t} = -S_{im} + \Gamma_w, \quad \theta = \theta_{mo} + \theta_{im} \end{aligned} \tag{8}$$

where subscripts *mo* and *im* refer to the mobile and immobile domains, respectively, θ_{mo} and θ_{im} are volumetric soil water contents in the mobile and immobile pore domains [L^3L^{-3}], h_{mo} is the pressure head in the mobile domain [L], K_{mo} is the hydraulic conductivity [LT^{-1}], S_{mo} and S_{im} are the sink terms [T^{-1}], and Γ_w is the mass exchange between the mobile and immobile regions [T^{-1}]. Equations 8 were solved using one set of soil water retention and hydraulic conductivity curves defined for the mobile domain and another soil water retention curve for the immobile domain. The van Genuchten Eqs. 4 and 5 were used to describe these soil hydraulic properties. The mass exchange between the mobile and immobile regions was calculated using the following equation:

$$\Gamma_w = \omega_w(h_{mo} - h_{im}) \tag{9}$$

where ω_w [$L^{-1}T^{-1}$] are first-order rate coefficients and h_{im} is the pressure head in the immobile domain [L].

The dual-porosity formulation for solute transport was based on the advection-dispersion and mass balance equations:

$$\begin{aligned} &\frac{\partial \theta_{mo} c_{mo}}{\partial t} + \frac{\partial f_s \rho_d s_{mo}}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\theta_{mo} D_{mo}\frac{\partial c_{mo}}{\partial r}\right) + \frac{\partial}{\partial z}\left(\theta_{mo} D_{mo}\frac{\partial c_{mo}}{\partial z}\right) - \left(\frac{\partial q_{mo} c_{mo}}{\partial r} + \frac{\partial q_{mo} c_{mo}}{\partial z}\right) - S_{c,mo} - \Gamma_s \\ &\frac{\partial \theta_{im} c_{im}}{\partial t} + \frac{\partial (1-f_s)\rho_d s_{im}}{\partial t} = -S_{c,im} + \Gamma_s \end{aligned} \tag{10}$$

where f_s is the dimensionless fraction of sorption sites in contact with the mobile water and Γ_s is the solute transfer rate between the two regions [$ML^{-3}T^{-1}$]. The solute transfer was described using the following equation:

$$\Gamma_s = \omega_s(c_{mo} - c_{im}) + \Gamma_w c^* \tag{11}$$

where ω_s [T^{-1}] is the first-order rate coefficient, and c^* is equal either to c_{mo} for $\Gamma_w>0$ or to c_{im} for $\Gamma_w<0$.

Simulation of water flow

Water flow within the radially symmetric soil samples was simulated using the HYDRUS-2D software. The 2D-transect (10×25 cm) representing the finite element mesh for half of the column and the applied boundary conditions (BC) is shown in Fig. 2. The boundaries and initial conditions were set according to the laboratory controlled and measured conditions. No water flux was defined for either lateral boundary. The seepage face was assumed at the bottom boundary (e.g. the water outflow was calculated only when the lower end of the soil column became saturated). The upper boundary was separated into two parts. Firstly, variable flux for the area with diameter of 2 cm (represented irrigation) was established. More precisely, in the case of the scenario C and during the first two weeks for the scenario V, the first day infiltration rate was 1500 $cm^3 day^{-1}$ and for the remaining six days there was no water flux. Then, in the case of scenario V, due to the high willow water demand the irrigation had to be applied 2-times per week (first and third day infiltration rate of 1500 $cm^3 day^{-1}$; remaining five days with no flux). Secondly, the rest of the upper boundary was represented by an atmospheric boundary (evaporation–E/transpiration–T), which was calculated using Eq. 8.

$$E/T = \frac{V_i - [V_o + V(\theta_{n+1} - \theta_n)]}{\pi(r^2 - r_i^2)t} \tag{12}$$

where E is evaporation [LT^{-1}], T is transpiration [LT^{-1}], V_i is inflow volume [L^3], V_o is outflow volume [L^3], V is volume of the soil in pot [L^3], θ_n and θ_{n+1} are measured volumetric water contents [L^3L^{-3}], r is radius of the soil column [L], r_i is radius of inflow area [L] and t is time in days [T]. Evaporation was applied in the case of scenario C, whereas in the case of scenario V the evaporation was disregarded since the soil top was covered by gravel (reducing evaporation) and plant transpiration was considered instead. No root growth and a homogenous root distribution within the soil was assumed (Eq. 3; $b=1/V_c$, where V_c is the volume of the column) because a homogenous root distribution in the soil column was anticipated during the second vegetation period. The actual

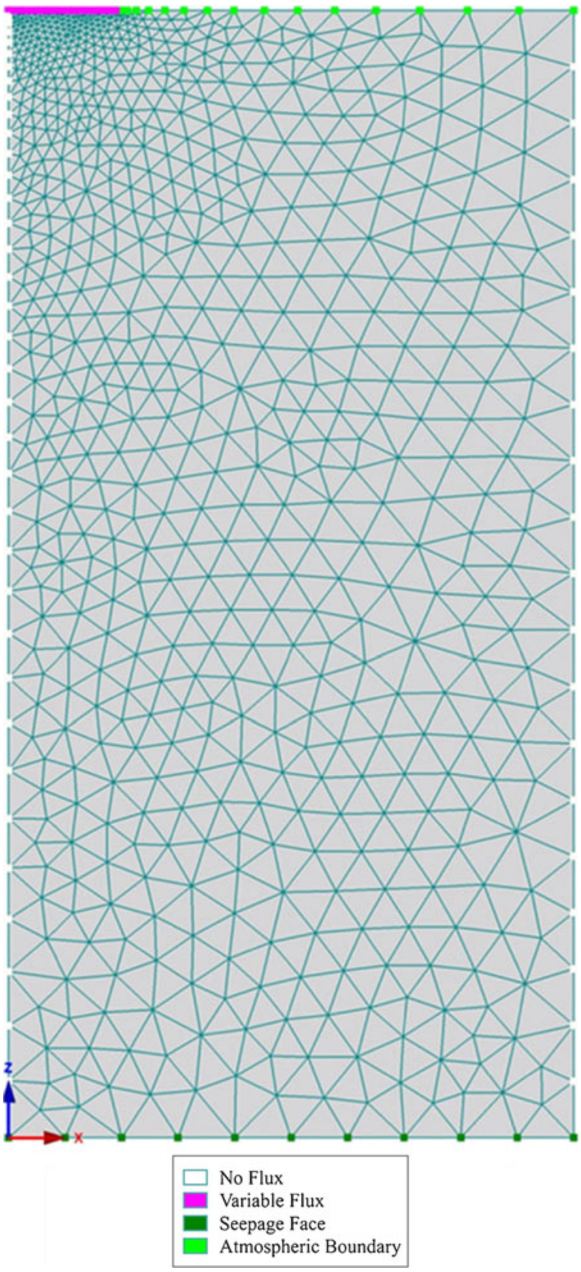

Fig. 2 Finite element mesh and boundary conditions of the radially symmetric flow domain

transpiration calculated using Eq. 12 was set instead of the potential transpiration. Therefore, the parameters of the alpha-function were adjusted to ensure unreduced water uptake by plants (e.g. the value of alpha-function always equalled 1). The initial conditions were defined as the initial volumetric water content θ_0 (average values within the soil column were measured using FDR probe).

Parameters of the van Genuchten soil hydraulic functions were partly set to the measured values and partly estimated using the numerical inversion of transient flow data. The saturated volumetric water contents θ_s, obtained in the laboratory in 100-cm^3 soil samples (3×2 samples for each scenario) using a saturation pan, were used for simulations (Table 3). The difference between the saturated water contents for scenarios C and V corresponded to the measured root volume which was, on average, 0.047 [cm^{3}cm^{-3}]. The residual volumetric water contents θ_r, were set to low values. Remaining parameters (α, n and K_s) were numerically optimized using the measured water contents (average values within the soil column) and cumulative outflows at the soil columns bottoms over a period of time using the single-porosity model in HYDRUS-2D. The following objective function Φ was minimized (13):

$$\Phi(d,q,p) = \sum_{j=1}^{m_q} v_j \sum_{i=1}^{n_{qj}} w_{i,j}\left[q_j^*(x,t_i) - q_j(x,t_i,d)\right]^2 \tag{13}$$

where the term on the right-hand side represents deviations between the measured and calculated space-time variables (cumulative water fluxes across a bottom boundary versus time, average soil water content within the soil column versus time), m_q is the number of different sets of measurements, n_{qj} is the number of measurements in a particular measurement set, q_j*(x,t_i) represents specific measurements at time t_i, for the j-th measurement set at location $x(r,z)$ and $q_j(x, t_i, d)$ are the corresponding model predictions for the vector of optimized parameters, d (e.g., α, n or K_s), v_j and $w_{i,j}$ are weights associated with a particular measurement set or point.

Parameter optimization was first carried out with the assumption of isotropic soil media (e.g. $K_{rr}{}^A$ and $K_{zz}{}^A$ anisotropic factors equalled 1) for both scenarios (with and without plants). Since the α and n values for soil with plant roots were very different in comparison to the soil without the roots (see the discussion section), the anisotropy of soil caused by the vertical root orientation was assumed. Three simulations were performed. While the anisotropy factor $K_{rr}{}^A$ always equalled 1, the anisotropy factor $K_{zz}{}^A$ equalled 10, 15 and 20, respectively. Only results for $K_{zz}{}^A$=1 and 10 are shown here.

Then, the dual-porosity model was also applied to approximate probable non-equilibrium water flow. Since

Table 3 Parameters of soil hydraulic functions (van Genuchten 1980), values of the minimized objective function and R^2 describing the relationship between simulated and measured cumulative outflow at the soil column bottom and average soil water content for C vs. all V scenarios (K_{zz}^A – anisotropy factor/tensor in the vertical direction). Parameters of dual-porosity model are shown for mobile/immobile soil media

Scenario		θ_r	θ_s	α	n	K_s	Objective function		R^2
		[$cm^3 cm^{-3}$]	[$cm^3 cm^{-3}$]	[cm^{-1}]	[−]	[$cm day^{-1}$]	0[c]	2[c]	
Control		0.05[a]	0.43[a]	0.032±0.002[b]	1.53±0.02[b]	29.1±0.3[b]	0.005	1.216	0.999
Vegetation	(anisotropy $K_{zz}^A=1$)	0.05[a]	0.38[a]	0.06[a]	3.50[a]	3208±0.01[b]	0. 249	9.150	0.992
	(anisotropy $K_{zz}^A=10$)	0.05[a]	0.38[a]	0.049±0.001[b]	2.78±0.08[b]	33.6±0.67[b]	0.204	12.10	0.987
	(dual-porosity–mobile)	0.00[a]	0.23[a]	0.040[a]	2.54±0.20[b]	17.9±1.28[b]	0.032	–	0.996
	(dual-porosity–immobile)	0.05[a]	0.15[a]	0.015[a]	1.50[a]	–	–		–

[a] Set values

[b] 95% confidence limits of optimized soil hydraulic parameters

[c] The value of the objective function (0—measured volumetric water content - average through soil profile; 2—Cumulative boundary fluxes across a bottom boundary = cumulative seepage face)

the experiment did not provide enough data to optimize all parameters, many of them were set to values estimated based on previous studies and examples listed in HYDRUS programs (Radcliffe and Šimůnek 2010). The parameters of the soil water retention curve in the immobile ($\theta_{s,im}$, $\theta_{r,im}$, α_{im} and n_{im}) and mobile ($\theta_{s,mo}$, $\theta_{r,mo}$ and n_{mo}) domain (Table 3) were set as follows: saturated soil-water content in the mobile and immobile domain equaled to 0.23 and 0.15 $cm^3 cm^{-3}$, respectively, (which was based on the soil structure observation) $\theta_{r,mo}$ in the mobile domain was zero, $\theta_{r,im}$ equaled to the θ_r for equilibrium water flow, and the greater retention ability in the immobile domain in comparison to the mobile domain was assumed and characterized by lower values of α_{im} and n_{im} in the immobile domain. Remaining parameters (α_{mo}, $K_{s,mo}$ and ω_w) were numerically optimized using the cumulative outflows at the soil columns bottoms over a period of time using the dual-porosity model in HYDRUS-2D.

Simulation of metal transport

Soil hydraulic parameters resulting from the analysis of water flow were then used as input data for direct simulation of the metals transport. The same finite element mesh, boundary and initial conditions for water flow were used as before. The following boundary conditions were applied for the metals transport simulation (no flux at both lateral boundaries and a third type boundary condition at the top and bottom boundary). Initial metal concentrations were defined according to the measured (by ICP-OES) concentrations in soil water extracted from the soil, using suction cups (Table 2). The bulk density ρ_d was measured in the 100-cm^3 soil samples (Table 4).

The parameters of the Freundlich adsorption isotherms were previously measured and determined by Trakal et al. (2012). In order to evaluate the sorption behaviour of chosen metals, a batch sorption experiment was performed. Metal sorption, represented by K_F values decreased in the order $K_F(\mathrm{Pb}) > K_F(\mathrm{Cu}) >> K_F(\mathrm{Zn}) > K_F(\mathrm{Cd})$. Furthermore, sorption parameters (Table 4) showed that root presence and production of the root exudates had a negligible effect on the metal sorption behaviour (Trakal et al. 2012). Additionally, a new batch experiment was performed according to the methodology described in Trakal et al. (2012). Citrate (representing a root exudate involved in metal mobilization processes) was considered during the determination of Freundlich parameters in scenario V. More precisely, 0.5 and 5 mM citric acid was mixed with metals solution in order to evaluate its effect on the final Freundlich parameters (Poulsen and Hansen 2000).

Measured metal concentrations did not provide enough information for optimizing values of longitudinal and transversal dispersivities; therefore the values of 5.0 and 0.5 cm were set as the longitudinal and transversal dispersivity, respectively, which was estimated based on the studies performed by Kodešová et al. (2009) on undisturbed soil columns and by Kodešová et al. (2012a) on repacked soil columns.

The solute transfer between the mobile and immobile domains (when using the dual-porosity model) due to the water transfer was only assumed.

The metal root uptake was simulated in the case of the scenario V. The maximum concentration (the limited value of c_{Root}), which can be removed from the soil due to the root uptake was adjusted manually when necessary. Adjustment was performed by comparing the total quantity of metals accumulated in the top part of the plant measured at the end of the second vegetation period and the simulated cumulative solute uptake.

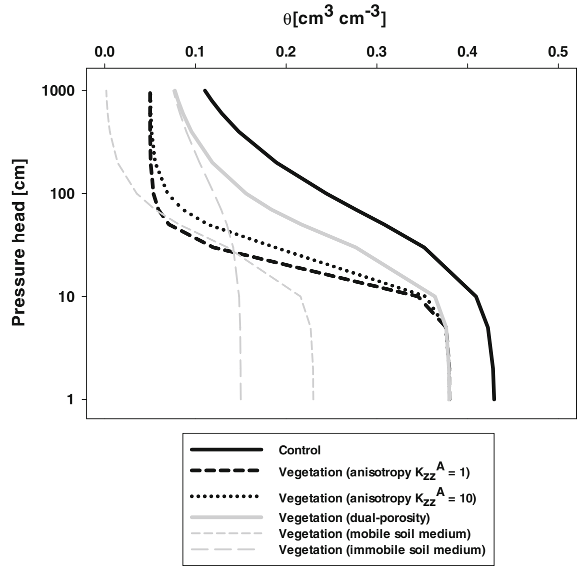

Fig. 3 Soil water retention curves for C and all V scenarios (K_{zz}^{A} – anisotropy factor/tensor in the vertical direction). Dual-porosity model was also considered for V scenario (retention curves for mobile/immobile mediums are shown in the figure)

Results

Water flux

Transient water flow data (cumulative outflow at the soil column bottom and average soil water contents) measured during the second vegetation period were used for calibrating the radially symmetric HYDRUS model. The van Genuchten hydraulic parameters, which were optimized, are shown in Table 3. Results demonstrate reliably estimated parameters in the case of the scenario C (when isotropic media was assumed), which is indicated by their narrow 95 % confidence limits, low values of the objective functions and a high value of the coefficient of determination R^2. On the other hand, the parameter optimization for the scenario V (when assuming the isotropic media) led to very high values of α and n, which caused numerical instability. Therefore, the α and n values were fixed at 0.06 cm^{-1} and 3.50, respectively, and K_s value was only estimated. In this case, the 95 % confidence limits for K_s were again very narrow, but values of the objective functions were significantly higher and the coefficient of determination R^2 was lower in comparison to those from the C scenario. In addition, the shape of the retention curve was very different in comparison to the shape of the retention curve for the C scenario (Fig. 3). The optimized K_s value from the V scenario was 100 times higher than that from the value from the C scenario. The soil

Table 4 Parameters for metal simulations: bulk density (ρ_d; data shown are means±SD (n=3)), longitudinal (D_L) and (D_T) transversal dispersivity, parameters of the Freundlich adsorption isotherm (K_F, β)

Scenario		ρ_d	D_L[a]	D_T[a]	K_F[b] [$cm^{3\beta}$ $nmol^{1-\beta}$ g^{-1}]				β[b] [−]			
		[gcm^{-3}]	[cm]	[cm]	Cd	Cu	Pb	Zn	Cd	Cu	Pb	Zn
Control		1.43±0.04	5.00	0.50	51.0	234	337	74.5	0.245	0.276	0.264	0.230
Vegetation[c]	0 mM CA	1.43±0.04	5.00	0.50	51.6	246	358	87.6	0.249	0.282	0.275	0.256
	0.5 mM CA	1.43±0.04	5.00	0.50	64.9	255	372	139	0.265	0.229	0.313	0.230
	5.0 mM CA	1.43±0.04	5.00	0.50	70.3	172	187	86.7	0.317	0.156	0.412	0.405

[a] Value of D_L and D_T were obtained from Kodešová et al. (2009, 2012a)

[b] Freundlich adsorption isotherm parameters K_F, β were obtained from Trakal et al. (2012)

[c] Represented all 3 V scenarios (single-porosity with anisotropy K_{zz}^{A} =1, 10 and dual-porosity soil mediums)

CA citric acid

parameters for scenario V were similar to sandy material, which did not seem to be realistic. Since the very fast reaction of the soil porous system was monitored (rapid infiltration at the soil column top and almost immediate discharge at the bottom), the anisotropy of soil caused by the vertical root orientation was assumed (soil structure compression in the horizontal direction and thus vertical flow predisposition). The anisotropy factor K_{rr}^{A} always equalled 1, the anisotropy factor K_{zz}^{A} equalled 10, 15 and 20, respectively. The resulting parameters for one simulation ($K_{zz}^{A}=10$) are shown in Table 3. It should be noted that the hydraulic conductivity in the horizontal direction equals the value in the table and the hydraulic conductivity in the vertical direction equals the value in the table multiplied by the anisotropy factor. The shape of the retention curve for $K_{zz}^{A}=10$ is shown in Fig. 3. In all cases, the 95 % confidence limits were very narrow, but values of the objective functions were again higher and the coefficient of the determination R^2 was lower in comparison to those from the C scenario. However, the shape of the soil hydraulic characteristics changed significantly. The slope of the retention curve increased with the increasing anisotropy factor K_{zz}^{A}. The K_s values for all three scenarios were close to the K_s value for the C scenario. Next, the non-equilibrium water flow was assumed and the dual-porosity model was applied to simulate observed fast response of the soil porous system. The 95 % confidence limits were slightly wider then for the previous simulations, but the value of the objective function (calculated for outflow data) decreased and the coefficient of determination R^2 (calculated for outflow data only) increased (Table 3). The retention curves for both domains and their sum (characterizing retention curve of the entire flow region) are shown in Fig. 3. It is evident that the shape of the retention curve for the entire flow domain is similar to that of the C scenario. The $K_{s,mo}$ value is even slightly lower than K_s value for the C scenario (Table 3). The relatively low value of first-order rate coefficients ω_w (Table 3; e.g. slow interaction between the mobile and immobile domains) was obtained.

Table 5 and Fig. 4 show the measured and simulated boundary fluxes (negative values indicate an inflow into the modelled system). An excellent fitting of the modelled cumulative inflow and outflow in the scenario C was obtained. Whereas in the case of scenario V the curve modelled using the single-porosity model (not shown in Fig. 4) was slightly different in comparison with observed data. The reason was that the simulated soil water content was considerably higher than the monitored one (see below), as more water had remained in the soil. Both mismatches were caused by optimized values for the hydraulic parameters in the V column, which were estimated with lower reliability (as mentioned above). The model was unable to closely reproduce observed data, as indicated by the objective function values. The better correspondence between the measured and simulated outflows was obtained when using the dual-porosity model (Table 5; Fig. 4). The restricted water flux into the immobile domain was simulated due to the low value of ω_w. As a result, the mobile domain was quickly saturated during irrigation allowing fast water discharge at the soil column bottom. Figure 4 and Table 5 also show a little water discharge due to the evaporation from the column surface (scenario C) and 10-times higher water discharge due to the willows transpiration (scenario V), which were both observed in the laboratory and set as input data.

The modelled and measured data of the volumetric water content showed an excellent agreement in the scenario C (Fig. 5), but a weak correspondence of the measured and simulated (using the single-porosity model) data were obtained in both V scenarios.

Table 5 Simulated vs. measured water cumulative outflow, evaporation and transpiration for C and all V scenarios

Scenario	Cumulative outflow [l]				Cumulative evaporation [l]		Cumulative transpiration [l]	
	Measured	Modelled[a]			Measured	Modelled	Measured	Modelled
		Anisotropy ($K_{zz}^{A}=1$)	Anisotropy ($K_{zz}^{A}=10$)	Dual-porosity				
Control	12.2	11.8	–	–	0.94	0.94	–	–
Vegetation	12.5	10.3	10.9	12.7	–	–	11.6	11.6

[a] V scenario modelling with applied anisotropy factor/tensor in the vertical direction – K_{zz}^{A} and with dual-porosity model consideration

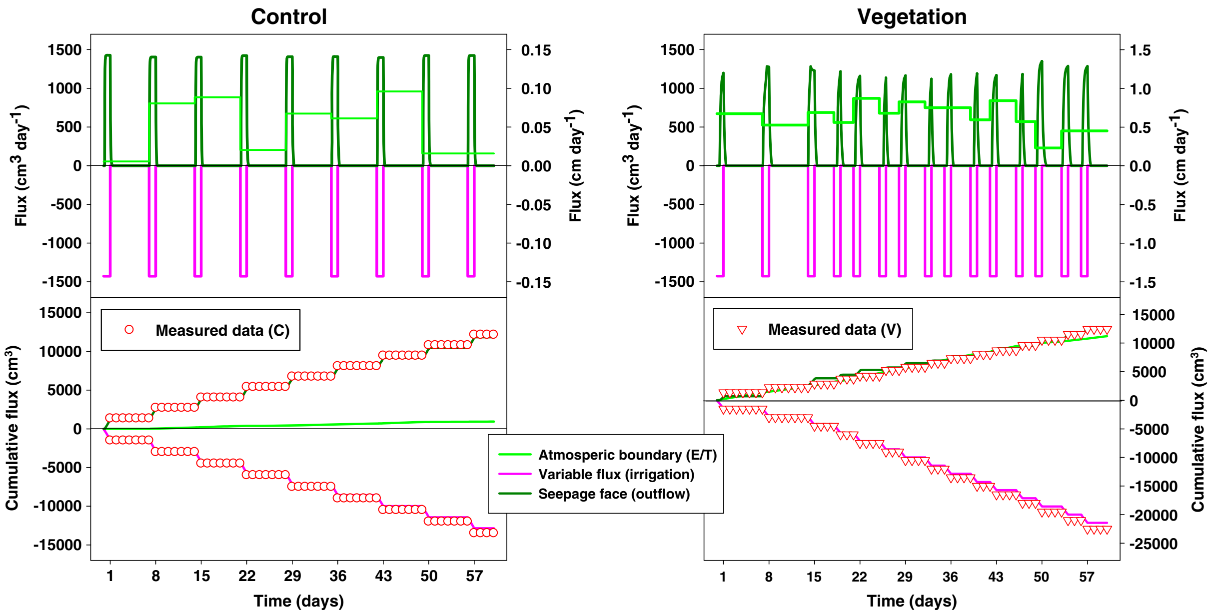

Fig. 4 Measured and simulated boundary fluxes and cumulative water fluxes of the C and V scenario (only dual-porosity consideration is shown in the figure). Fluxes, which are positive, are out of the flow domain, negative are into the flow domain

Specifically, for the first week of monitoring, the measured data were in agreement with the modelled ones, but the subsequently observed data were remarkably different in comparison to the simulated ones (a difference of about 0.13 $cm^3 cm^{-3}$; Fig. 5). However, the increasing/decreasing trends of the modelled and measured water content behaviour after/before irrigation

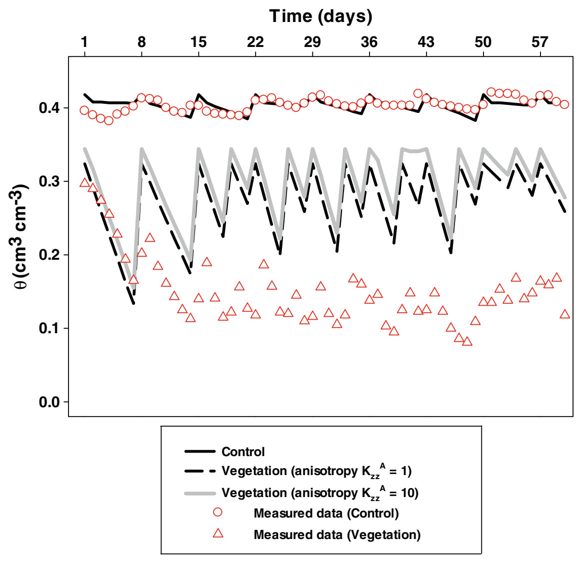

Fig. 5 Measured and simulated soil water contents of the C and all V scenarios (K_{zz}^A – anisotropy factor/tensor in the vertical direction; was considered for the V scenario)

were similar. The higher simulated water storage in the soil column in comparison to the measured one (which for $\Delta\theta$=0.13 $cm^3 cm^{-3}$ would be 1020 cm^3) corresponds to the lower simulated cumulative outflow compared to the measured one.

Additionally, and for illustration, Fig. 6 (documenting simulated water content distributions one day before irrigation, during irrigation, and one day after irrigation) shows how water may have infiltrated into the soil and how rapidly water may have redistributed within the soil column. Figure 6 also explains why water may have drained from the soil at the time of irrigation in both cases; when θ_s was reached on the bottom of the column (boundary seepage face). Figure 6 clearly show considerably different water distributions within the soil columns. While the soil water content near the column bottom is close to saturation for K_{zz}^A=1 for all 3 stages, the soil water content for K_{zz}^A=10 is lower and larger in the column before and after the irrigation, respectively. Figure 6 also shows greater saturation in the centre of the column during the infiltration for K_{zz}^A=10. In the case of the result of the dual-porosity model, the soil water contents in the mobile domain simulated during the irrigation are closer to saturation then water contents in the immobile domain as well as water contents simulated using the single-porosity model. Thus greater outflow was obtained as mentioned above.

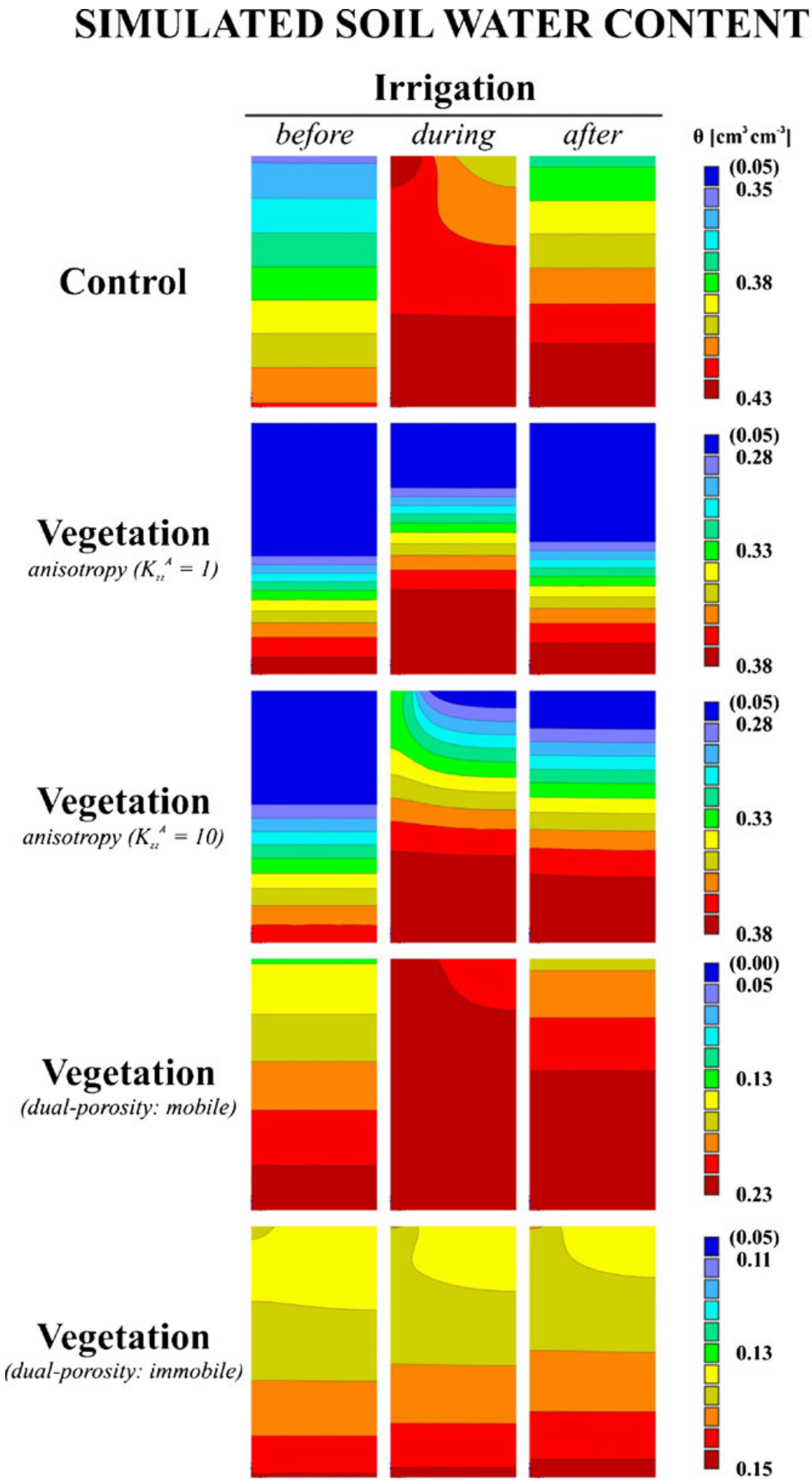

Fig. 6 Simulated soil water contents ($cm^3 cm^{-3}$) within the flow domain simulated in the radially symmetric flow domain (C vs. all V scenarios) one day before, during and one day after irrigation (situation, how infiltration may have occurred)

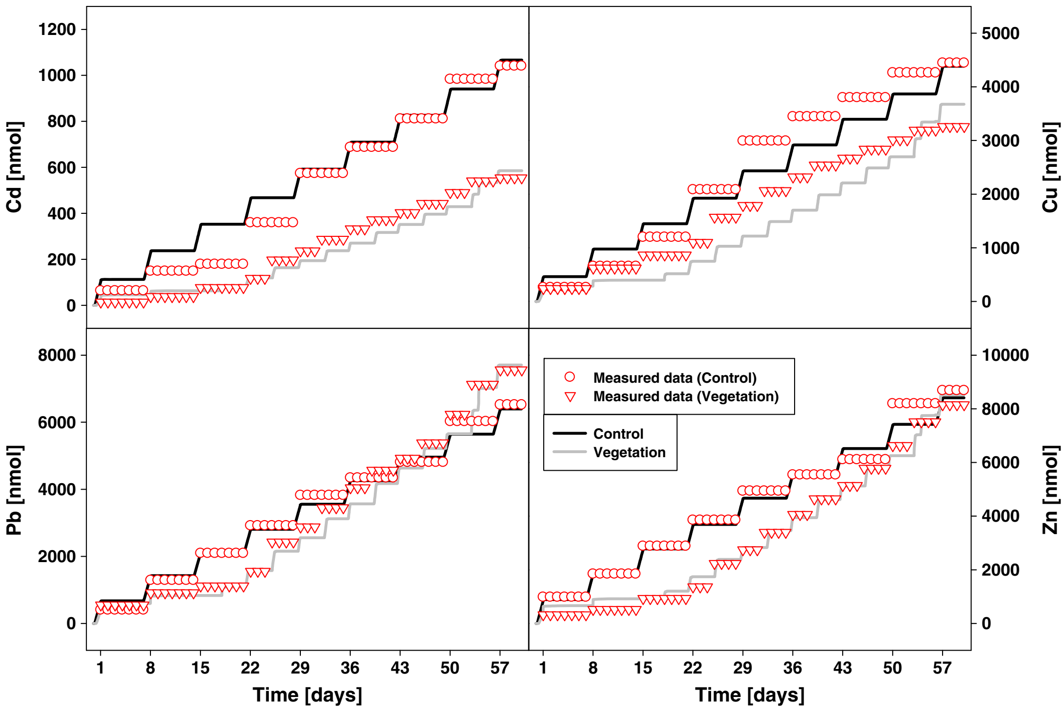

Fig. 7 Measured and simulated cumulative metal outflows (seepage face) of the C and V scenarios (only dual-porosity consideration is shown in the figure)

Metal transport

A modelled cumulative metal content in the leachate was in good agreement with observed data for all four metals in the scenario C (Fig. 7; Table 6). In the case of the scenario V the modelled metal cumulative outflow was reduced in comparison to the observed data for all metals (Table 6) when using the single-porosity

Table 6 Simulated vs. measured final cumulative metal outflow for C and all V scenarios

Scenario	Metal	Cumulative metal outflow [μmol]						
		Measured	Modelled[a]					
			Anisotropy (K_{zz}^{A}=1)		Anisotropy (K_{zz}^{A}=10)		Dual-porosity	
			A[b]	B[b]	A[b]	B[b]	A[b]	B[b]
Control	Cd	1.04	1.07	–	–	–	–	–
	Cu	4.44	4.38	–	–	–	–	–
	Pb	6.51	6.40	–	–	–	–	–
	Zn	8.68	8.41	–	–	–	–	–
Vegetation	Cd	0.57	0.51	0.48	0.53	0.51	0.59	0.61
	Cu	3.31	3.22	2.99	3.31	3.18	3.68	3.80
	Pb	7.70	6.76 (6.78)[c]	6.28 (6.30)[c]	6.93 (6.95)[c]	6.67 (6.69)[c]	7.71 (7.73)[c]	7.98 (8.01)[c]
	Zn	8.32	7.46	6.94	7.66	7.37	8.52	8.82

[a] V scenario modelling with applied anisotropy factor/tensor in the vertical direction – K_{zz}^{A} and dual-porosity model consideration

[b] A/B – cumulative metal transport without/with citrate consideration in case of V scenario

[c] Modified Pb uptake using maximal concentration, c_{root}=0.17 [nmol cm^{-3}]

model. This finding was explained by a reduction in the modelled water outflow compared with the measured one (Table 5). A slightly higher outflow of metals was simulated when assuming anisotropic conditions (K_{zz}^{A}=10) which was in agreement with the larger outflow of water (Table 5). Furthermore, in the case of the dual-porosity soil medium simulated cumulative metal outflow showed the best fit to the measured data (Fig. 7; Table 6). In general, a comparison of the V and C scenarios showed differences for modelled and measured metal cumulative outflows. In the case of Pb, no significant changes were observed. More precisely, in comparison with the C scenario the cumulative Cd outflow decreased approximately 2-times and the cumulative Cu and Zn outflow showed an almost 20 % reduction for the V scenario. These results were due to the lower initial metal concentration used in the case of the V scenario (Table 2) and also due to the more intensive irrigation caused by willow transpiration (Fig. 4). Additionally, the modelled as well as the measured cumulative metal outflows were reduced in the case of the V scenario, which were also attributable to the root metal uptake, especially in case of Cd and Zn.

The data of the metals accumulation in willow tissues from this work have been partly used. In particular, the total metal content in the aboveground biomass after the second vegetation period was compared with the modelled ones. In more detail, the final measured Cd and Zn content in the aboveground biomass after the second vegetation period was 12.5 and 84.5 [μmol], respectively. On the other hand, the modelled cumulative root Cd and Zn uptake into the willow aboveground tissues was 0.51 and 7.59 [μmol] (Table 7); hence the modelled Cd and Zn uptake was 20- and 10-fold lower, respectively. The modelled cumulative Cu root uptake of 3.27 [μmol] was almost in agreement with the measured content in the aboveground biomass 4.01 [μmol]. Finally, the modelled root Pb uptake showed a higher value compared with the observed one (Table 7), hence the modelled uptake was modified. A maximum concentration c_{root} of 0.17 [nmol cm^{-3}] for Pb uptake was set. This requirement was used due to (i) extremely low Pb translocation from the experimental soil into the willows (Kos et al. 2003); and (ii) a rather high initial Pb concentration. Soil anisotropy had no effect on the root metal uptake for all metals (Table 7). Additionally, the measured total metal content in the roots was not taken into account because of a hypothetical minimum root growth and the consequent anticipated negligible metals accumulation during the second vegetation period. Hence, for the comparison with the modelled cumulative metal uptake only the total metal content in top willow parts was used. Nevertheless, for illustration, the total metal content in the roots as a sum of both vegetation periods was as follows Cd=4.70 μmol; Cu=6.71 μmol; Pb=54.6 μmol; and Zn=30.9 μmol. Finally, the citrate consideration had negligible effect on simulated cumulative metal leachate and metal uptake for all three V scenarios (Tables 6 and 7).

Figure 8 shows the simulated final Cd, Cu, Pb and Zn concentration distribution within the soil column

Table 7 Simulated vs. measured final cumulative metal uptake for all V scenarios (anisotropy K_{zz}^{A}=1, 10 and dual-porosity model calculation)

Scenario	Metal	Cumulative metal uptake [μmol]						
		Measured	Modelled[a]					
			Anisotropy K_{zz}^{A}=1		Anisotropy K_{zz}^{A}=10		Dual-porosity	
			A[c]	B[c]	A[c]	B[c]	A[c]	B[c]
Vegetation	Cd	12.5	0.52	0.52	0.51	0.52	0.50	0.50
	Cu	4.01	3.24	3.21	3.23	3.21	3.17	3.14
	Pb	1.89	6.78 (1.89)[b]	6.76 (1.89)[b]	6.78 (1.89)[b]	6.76 (1.89)[b]	6.63 (1.89)[b]	6.61 (1.89)[b]
	Zn	84.5	7.33	7.39	7.33	7.39	7.17	7.23

[a] V scenario modelling with applied anisotropy factor/tensor in the vertical direction – K_{zz}^{A} and dual-porosity model consideration

[b] Modified Pb uptake using maximal concentration, c_{root}=0.17 [nmol cm^{-3}]

[c] A/B – cumulative metal transport without/with citrate consideration in case of V scenario

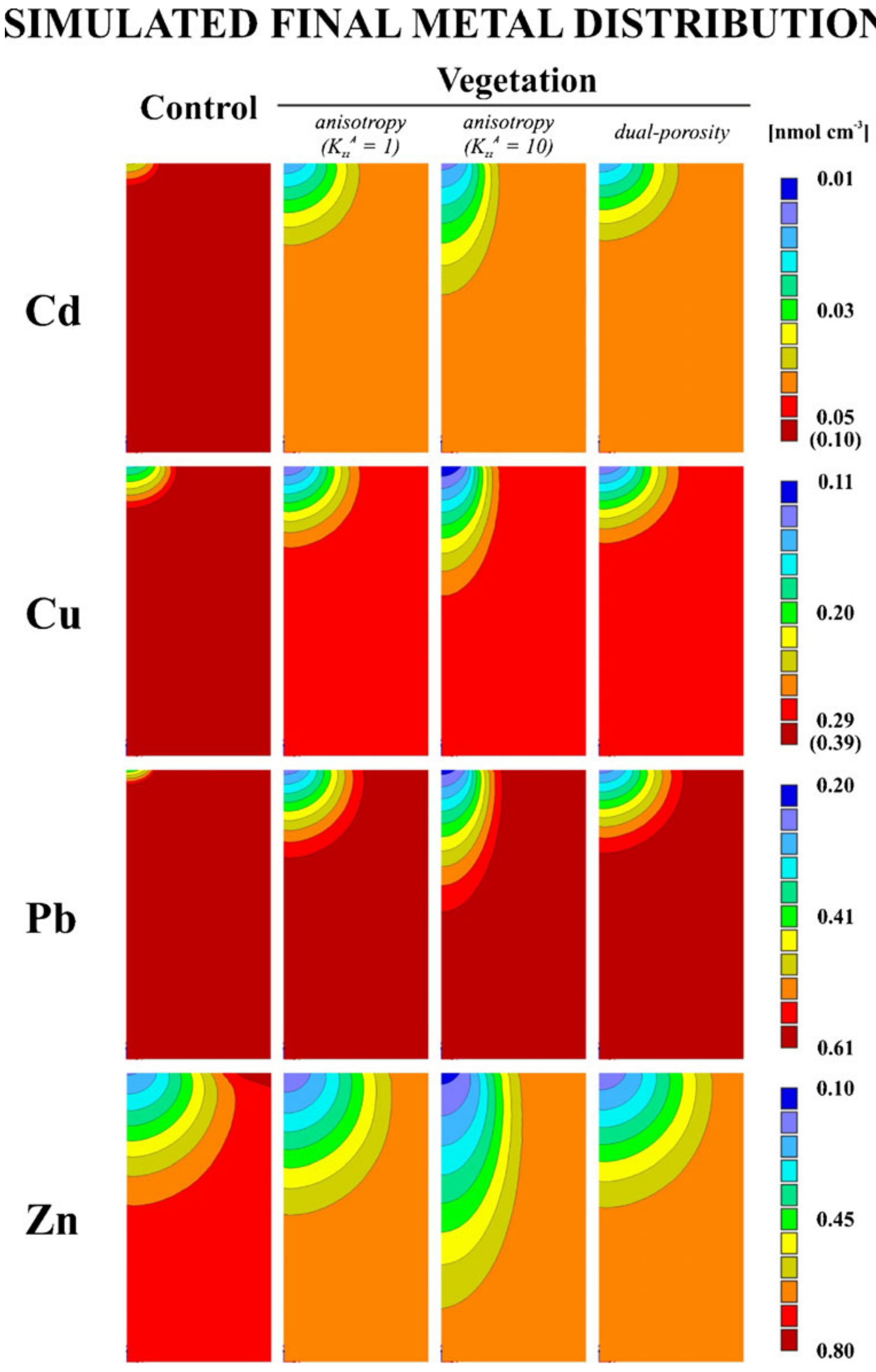

Fig. 8 Final metal concentration (nmol cm^{-3}) within the soil profile simulated in two-dimensional radial symmetric model (C vs. all V scenarios)

(not experimentally confirmed) for all scenarios (C, V for K_{zz}^{A}=1, 10 and dual-porosity soil media). The final Cd, Cu and Zn concentration distribution was affected in all V scenarios in comparison with the scenario C caused the most likely by: (i) lower initial metal concentration used for the V scenario (result of accumulation during first vegetation period); (ii) higher irrigation during the experiment for the V scenario (caused by evapotranspiration of willow); and (iii) root metal uptake (Table 7). In more detail, the final Cd and Zn concentration distribution for all V scenarios is the most intensively modified in comparison with the scenario C, mainly due to intensive Cd and Zn accumulation in willow tissues (Baum et al. 2006; Ingwersen et al. 2006; Leštan 2006) and also due to high Cd and Zn mobility in general. The final Cu and Pb concentration distribution was also modified, but not as intensively as for Cd and Zn. Additionally, applied anisotropy conditions (K_{zz}^{A}=10) showed an obvious effect on the final metal concentration distribution. On the other hand, dual-porosity soil media consideration showed minimal effect on final metal distribution in comparison with single-porosity isotropic model.

Discussion

Water flux

Water flux for both scenarios was simulated using the single-porosity model when isotropic media was considered for scenario C. Nevertheless, the anisotropy of soil caused by the vertical root orientation was assumed for V scenario mainly due to soil structure compression in the horizontal direction and thus vertical flow predisposition, which could be enhanced by root volume oscillation caused by oscillating soil water content; Hillel 1998). It was shown that the model was able into some degree approximate the observed data. However, the fast response of the soil system in the V scenario may be also attributed to the preferential flow. On the one hand, the great volume of the roots compressed the soil structure (which is indicated by the lower value of the measured saturated soil water content). On the other hand, large capillary pores (macro pores), e.g., preferential pathways, could be created. The roots' volume may have varied due to their water content oscillation, which corresponded to the soil water content changes (Hillel 1998). Thus a preferential pathway could be created along the roots (Köhne et al. 2009; Kodešová et al. 2011, 2012b). In such case a dual-permeability model (Gerke and van Genuchten 1993) in HYDRUS-2D would be more suitable to approximate the observed data (Kodešová et al. 2010). Nevertheless, this model was not used due to an insufficient quantity of the input data for the inverse calculation. Many parameters would have to be optimized or determined independently. In addition, dual system geometry would have to be studied using different imaging testing (NMR imaging, x-ray computer tomography, dye tracer imaging; Köhne et al. 2009) in different scales (Kodešová 2009, Kodešová et al. 2012b). Such analysis could not be performed since the main concern of our study was metal analysis in the soil and plants. Since measured data likely indicated the non-equilibrium water flow, the dual-porosity model (requiring fewer parameters) was at least applied to simulate the observed outflow. Reduced water transfer into to the immobile domain resulted into enhanced water flow in the mobile domain and thus in better correspondence between measured and simulated values.

Metal transport

Only the water soluble portion of metals (sampled by suction cups; Table 2) was considered for the calculation of root metal uptake into the experimental willows using the HYDRUS-2D model. However, in the case of Cd and Zn, the planted willows were able to accumulate these elements not only from the water soluble portion (Lasat 2002; Tlustoš et al. 2007) but also from the exchangeable portion. This has probably resulted from the production of root exudates leading to increased solubility and bioavailability of Cd and Zn (Fischerová et al. 2006). In more detail, Östman et al. (2008); Száková et al. (2010) showed that less than 0.5 % Cd was water soluble from the total metal content in the studied soil. On the other hand, Tlustoš et al. (2007) and Wieshammer et al. (2007) showed that the willow species used were able to accumulate 10 % from the total Cd during one vegetation period. This finding, therefore, explained the 20-times reduction in the modelled cumulative Cd uptake compared with the measured one. The same problem was found for Zn, but the reduction in the modelled cumulative Zn uptake was only 10-times lower, more probably due to the 0.1 % water soluble portion from the total

Zn content in the soil (Száková et al. 2008); and due to accumulation of 1 % from the total Zn content into the willow tissues (Tlustoš et al. 2007; Wieshammer et al. 2007). Moreover, modelled and measured root Cu uptakes were in good agreement, most likely due to the Cu uptake predominantly from the water soluble portion (Yang et al., 2005). Finally, simulated root Pb uptake was modified (using maximum concentration for Pb uptake) due to predominate Pb accumulation in the root zone (Garbisu and Alkorta 2001; Kos et al. 2003). Moreover, consideration of citrate presence in scenario V had negligible effect on cumulative outflow and root uptake for all metals main probably due to only partial affecting of Freundlich sorption parameters for all studied metals.

It should also be mentioned, that the root uptake of Ca and K (Nowack et al. 2006); P and As (Szegedi et al. 2010) was successfully modelled by the authors using the PHREEQC code as well as for the Cu uptake (Seuntjens et al. 2004) apparently due to calculations with the (bio)geochemical processes in more detail. Hence coupling the HYDRUS software package with a speciation/solubility geochemical model (e.g., PHREEQC-2) would provide more detailed and accurate information. Therefore, the HP1 program (coupling HYDRUS-1D with PHREEQC-2; Jacques et al. 2008; Šimůnek et al. 2009) would possibly be more suitable for modelling the root Cd, Cu, Pb and Zn uptake. However, this model could not be used for this specific two-dimensional radially symmetric problem. Additionally, for more realistic modelling of metal accumulation in roots (mediated by transporter proteins) uptake kinetics (by the Michaelis-Menten equation; Puschenreiter et al. 2005) should be also considered for the final simulation. Nevertheless, this fact should not be implemented for this simulation mainly due to no way to consider these kinetic parameters in the HYDRUS-2D model (as input data).

Conclusion

The major conclusions of this study regarding an optimization of soil hydraulic parameters; metal transport through the soil profile and consequent root uptake of metals into the experimental willows in the column experiment are as follows:

- Numerical optimization of the soil hydraulic parameters for a soil without (scenario C) and with experimental willows (scenario V) showed considerable differences between the soil hydraulic properties. The soil-porous system of the scenario C could be characterized as isotropic single-porosity soil media. In case of the scenario V, experimental results indicated that vertical roots modified the soil structure, therefore anisotropic single-porosity or dual-porosity soil media were considered.
- Simulated cumulative water and metal (Cd, Cu, Pb and Zn) outflow from the soil column in scenario C moderately approximated the measured data. The modelled cumulative water and metal outflow for scenario V increased when the anisotropic single-porosity or dual-porosity soil media were considered compared to the isotropic single-porosity system. The results of the dual-porosity model showed the best correspondence to the measured data.
- The measured cumulative Cd and Zn root uptake showed a 20- and 10-fold increased accumulation respectively in comparison with the model, due to the accumulation not only from the water soluble fraction (used for HYDRUS-2D calculations). The modelled Pb uptake was modified by reducing the maximum value of the root uptake concentration c_{Root} due to a very low accumulation and relatively high water soluble concentration. No significant impact of anisotropy or dual-porosity soil media consideration has been revealed.

Acknowledgments The experiment was financed from Norway funds (grant number CZ0092), Czech Ministry of Education (MSM6046070901) and Czech Science Foundation (GAČR P503/11/0840). The authors acknowledge namely Száková J, Nikodem A, Drábek O, Tejnecký V and Malík J (all from Faculty of Agrobiology, Food and Natural Resources, Czech University of Life Sciences Prague) for all soil/plant analyses and data evaluation. Additionally, the authors also acknowledge Mr. Alan M. Westcott for English corrections of the manuscript.

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